Lab Exercise 17- Creating Secrets in Kubernetes

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Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssword'
```

```
PS C:\Users\ASUS> kubectl create secret generic my-secret --from-literal=use
rname=admin --from-literal=password='P@ssw0rd'
secret/my-secret created
PS C:\Users\ASUS>
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

```
PS C:\Users\ASUS> kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-05T08:54:18Z"
  name: my-secret
  namespace: default
  resourceVersion: "21046"
  uid: c348bc81-1df0-4c41-baf0-7fd9302585e7
type: Opaque
PS C:\Users\ASUS>
```

The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssword” | base64
```

```
PS C:\Users\ASUS> [Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes("
P@ssw0rd"))
UEBzc3cwcmQ=
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt



echo -n 'my-password' > ./password.txt
```

```
PS C:\Users\ASUS> echo my-username > username.txt
PS C:\Users\ASUS> echo my-password > password.txt
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

```
PS C:\Users\ASUS> kubectl create secret generic my-file-secret --from-file=u
sername=./username.txt --from-file=password=./password.txt
secret/my-file-secret created
PS C:\Users\ASUS>
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
PS C:\Users\ASUS> kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: //5tAHkALQBwAGEAcwBzAHcAbwByAGQADQAKAA==
  username: //5tAHkALQB1AHMAZQByAG4AYQBtAGUADQAKAA==
kind: Secret
metadata:
  creationTimestamp: "2026-03-05T08:57:35Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "21209"
  uid: 4e98a231-78f7-40a8-b523-1f3ca2c555ea
type: Opaque
PS C:\Users\ASUS>
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
```

```
PS C:\Users\ASUS> kubectl create secret docker-registry my-registry-secret --docker-username=testuser --docker-password=testpass --docker-email=test@example.com --docker-server=docker.io
secret/my-registry-secret created
PS C:\Users\ASUS>
```

Verify the secret:

```
kubectl get secrets my-registry-secret -o yaml
```

```
PS C:\Users\ASUS> kubectl get secrets my-registry-secret -o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJkb2NrZXIuaW8iOnsidXNlcm5hbWUiOiJ0ZXN0dXNlciIsInBhc3N3b3JkIjoidGVzdHBhc3MiLCJlbWFpbCI6InRlc3RAZXhhbXBsZS5jb20iLCJhdXRoIjoiZEdWemRIVnpaWEk2ZEdWemRIQmhjM009In19fQ==
kind: Secret
metadata:
  creationTimestamp: "2026-03-05T09:00:07Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "21336"
  uid: faf7d30a-c449-4330-9eae-708969ef2326
type: kubernetes.io/dockerconfigjson
PS C:\Users\ASUS>
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
```

```
password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssword'
```

```
File    Edit    View

apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=
  password: UEBzc3cwcmQ=
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

```
PS C:\Users\ASUS> notepad my-manifest-secret.yaml
PS C:\Users\ASUS> kubectl apply -f my-manifest-secret.yaml
secret/my-manifest-secret created
PS C:\Users\ASUS>
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

```
PS C:\Users\ASUS> kubectl get secrets my-manifest-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","data":{"password":"UEBzc3cwcmQ=","username":"YWRta
W4="},"kind":"Secret","metadata":{"annotations":{},"name":"my-manifest-secre
t","namespace":"default"},"type":"Opaque"}
  creationTimestamp: "2026-03-05T09:01:00Z"
  name: my-manifest-secret
  namespace: default
  resourceVersion: "21380"
  uid: afebe859-bbf7-463e-8027-f04947474cca
type: Opaque
PS C:\Users\ASUS>
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
```

```
        key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

```
File    Edit    View

apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

```
PS C:\Users\ASUS> notepad pod-with-secret.yaml
PS C:\Users\ASUS> kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created
PS C:\Users\ASUS>
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

Step 7: Clean Up

Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
PS C:\Users\ASUS> kubectl delete secret my-secret
secret "my-secret" deleted from default namespace
PS C:\Users\ASUS> kubectl delete secret my-file-secret
secret "my-file-secret" deleted from default namespace
PS C:\Users\ASUS> kubectl delete secret my-registry-secret
secret "my-registry-secret" deleted from default namespace
PS C:\Users\ASUS> kubectl delete secret my-manifest-secret
secret "my-manifest-secret" deleted from default namespace
PS C:\Users\ASUS> kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted from default namespace
PS C:\Users\ASUS>
```